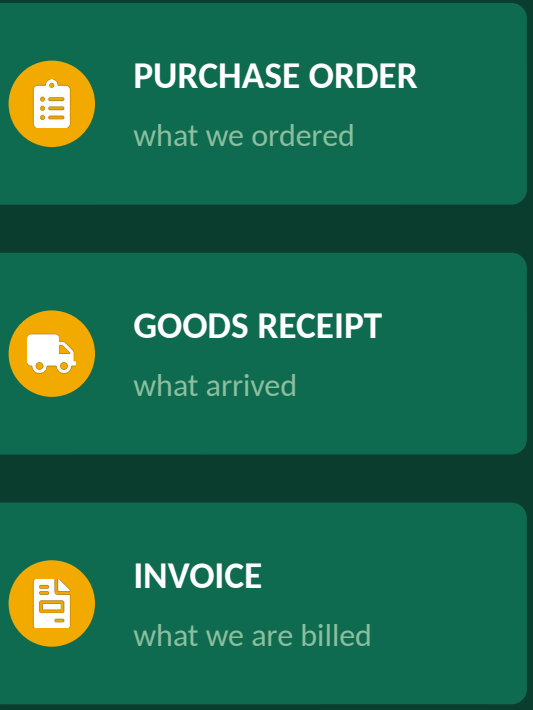
# Agentic AI for Accounts Payable

Invoice three-way match with a team of cooperating agents

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# Why invoices go wrong

Before paying a vendor, finance teams must check the invoice against the purchase order and the goods receipt. Done by hand, it is slow, and every miss becomes an overpayment.




**Domain:** Banking & Finance, Business Operations

## What slips through

- **Price above the PO**
- **Billed more than received**
- **Same invoice paid twice**

# From manual checks to agents

## Today: manual review

- Staff compare three documents by hand
- Slow and repetitive for every invoice
- Errors often found after payment
- No consistent written reason

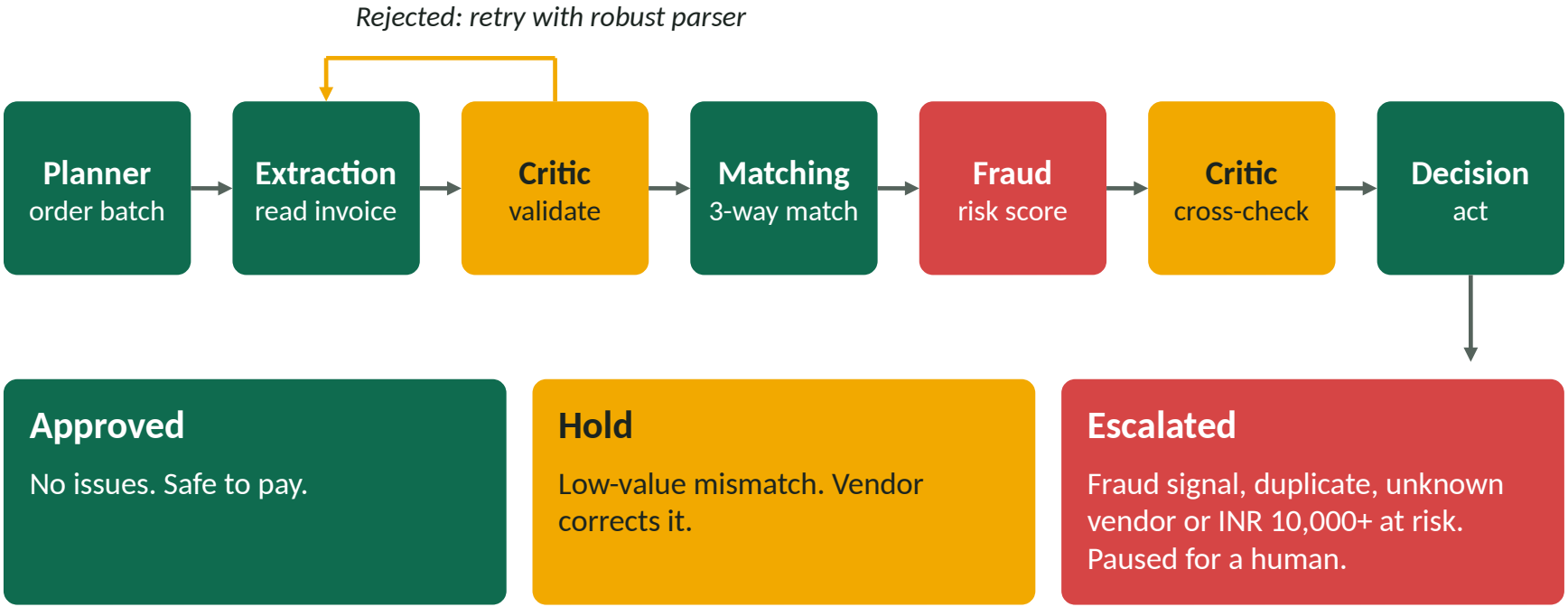
## With agents

- Planner orders the batch by urgency
- Agents extract, validate and match
- Problems caught before payment
- Every decision comes with a reason

# Seven agents, one job

| | |
|---|---|
| **Planner** | Sequences the batch by due date and amount; flags urgent invoices |
| **Extraction** | Turns invoice text into a typed schema with an LLM or parser; retries if rejected |
| **Critic** | Rejects bad arithmetic or invented values; gets a model second opinion; checks every variance |
| **Matching** | Compares invoice, PO and goods receipt on price, quantity, tax, vendor and duplicates |
| **Fraud** | Flags lookalike vendors, bank changes and spikes; scores and explains risk |
| **Decision** | Applies rules to approve, hold or escalate; writes the reason and a case summary |
| **Human Review** | Pauses escalations for a person, then updates vendor memory |

# How the agents hand off



# Fraud Agent: what matching misses

**+70**

pay-to bank account differs from the one on file

**+60**

vendor name one letter off an approved vendor

**+40**

total above 3x the vendor's largest past invoice

**+15**

suspiciously round total

- Risk score 0-100: these points, plus up to 20 for a vendor whose invoices were often escalated. 40 or more escalates.
- Every point comes with its reason, so the reviewer sees exactly why an invoice was stopped.
- The spike rule was tuned on 200,000 simulated invoices: about 0.02% false alarms.

# Human in the loop, with memory

**1 Pause**

LangGraph interrupt checkpoints the graph on escalation

**2 Review**

Read the case summary, then approve, hold or reject

**3 Resume**

The graph continues from its checkpoint

**4 Learn**

The decision is saved to vendor memory

**Real trace: a vendor's bank-account change, approved by a reviewer**

```
Fraud Agent: BANK_CHANGE; risk score 70/100
Decision Agent: ESCALATED (fraud / control risk: BANK_CHANGE)
... graph paused at interrupt; invoice waits in the Review queue ...
Human Review: Saafalya reviewed: APPROVE -> APPROVED - "new account confirmed by phone"
              new bank account saved to vendor memory: next invoice is not flagged
```

# The local model, with guardrails

**1 Extract**

Reads the invoice; every key field must appear in it

**2 2nd opinion**

Doubts trigger a retry, never an escalation

**3 Risk note**

Explains the rules' score in plain words

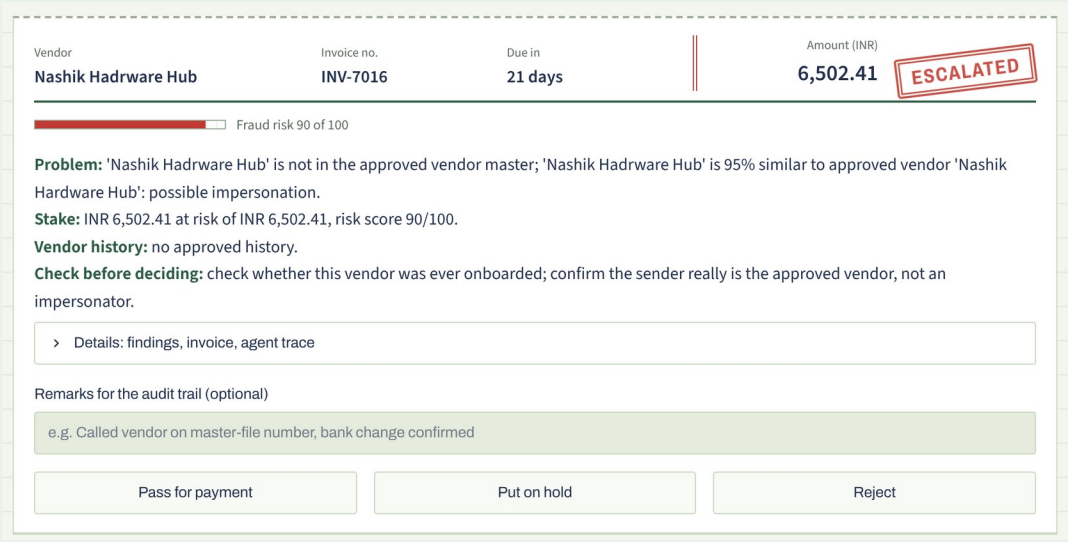
**4 Case brief**

Briefs the reviewer on each escalated invoice

**Real case summary for a bank-change invoice. Any number the model adds is rejected.**

```
Problem: pay-to account 50100990490278 differs from account on file 50100341316475.
Stake: INR 12,358.99 at risk, risk score 70/100.
Vendor history: 14 past approved invoices, largest INR 213,993.83.
Check: call the vendor on the number in the vendor master to confirm the new account.
```

# The reviewer's view




Review queue case card (real screenshot)

- Ledger-style app with rubber-stamp decisions and a seven-step pipeline strip
- Case summary up front; findings and trace one click away
- Approve, hold and reject carry equal weight
- Every decision goes to the audit trail and vendor memory

# System architecture

**Streamlit UI**
ledger-style app: live trace, pipeline strip, review queue, PDF upload, results

**LangGraph state graph**
shared state, retry edge, interrupt + checkpointer for human review

**Agents**

Planner | Extraction | Critic | Matching | Fraud | Decision | Human

**Tools and data**
PO DB, GRN DB, vendor master, duplicate registry, vendor memory, rules

**LLM**
Ollama local model (guarded) or rule-based mock

## Data flow

1. Batch or PDF enters
2. Graph shares state
3. Agents use tools and memory
4. Escalations pause for a human
5. Decision goes to the UI

# Demo walkthrough

**1 Generate batch**

24 synthetic invoices, 8 seeded errors, 3 of them fraud

**2 Run agents**

Watch the pipeline strip and live trace

**3 Review queue**

Read the case summary, then decide

**4 Upload PDF**

Run the agents on a PDF invoice

### Live trace for one invoice (real output)

```
Extraction Agent: attempt 1 via fast parser: INV-7009 | total 1.00
Critic Agent: extraction REJECTED (qty x unit != amount) -> retry
Extraction Agent: attempt 2 via robust parser: INV-7009 | total 1,734,135.13
Fraud Agent: AMOUNT_SPIKE (8.1x vendor's largest invoice); risk score 40/100
Decision Agent: ESCALATED (fraud risk score 40 >= 40) -> Review queue
```

# What the validation shows

**8/8**

seeded errors caught

**3/3**

fraud cases escalated

**0/16**

clean invoices wrongly flagged

**99.98%**

decided correctly across 37,200 generated invoices

- Default batch: 24 invoices, 8 seeded errors (3 fraud) and 16 clean; INR 0.99L at risk flagged, equal to the seeded total.
- The Critic loop fixed misread amounts before matching. 24 automated tests pass.
- High scores reflect rule-based matching on known data, not real-world accuracy.

# Limits and next steps

## Limitations

- Synthetic data and one fixed invoice layout
- Rule-based matching scores optimistically
- No OCR for scanned PDFs, no ERP link or login
- Time savings are estimates, not measurements
- Ollama runs are slower; small models can misread fields

## Future scope

- OCR for scanned PDFs and photos
- ERP connectors such as SAP and Tally
- Benchmark LLM extraction against the parser
- Learn fraud weights from reviewer decisions
- Role-based approvals and MCP tools

# Thank you

Questions?



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